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JOURNAL OF APPLIED CRYSTALLOGRAPHY

ISSN 1600-5767

Particle-size polydispersity analysis based on the unified exponential/power-law approach to small-angle scattering

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Received 10 November 2022
Accepted 30 May 2023

Edited by V. T. Forsyth, Lund University, Sweden, and Keele University, United Kingdom

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Keywords: polydispersity; size distribution; small-angle scattering; unified exponential/power-law approach; Porod integral; correlation lengths; hydrosols; nanoparticles.

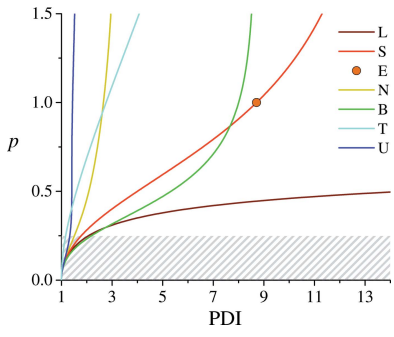
Controlling particle dispersity is of huge importance for practical applications in nanoscience and technology. The analysis of small-angle scattering of X-rays and neutrons for strongly polydisperse particulate systems is considered from the point of view of describing the type of size distribution function without applying classical regularization approaches. This article presents the development of a method for determining the polydispersity parameters of nanoobjects, based on the analysis of the ratio of various moments of the size distribution function, which are proportional to different invariants of the scattering curve. The use of the unified exponential/power-law approximation to describe small-angle scattering data makes it possible to determine the type of distribution, the average size and the spread. The possibilities of the method were tested for several hydrosols of metallic nanoparticles.

1. Introduction

The practical utility of nanosystems in various high-tech areas is now beyond any doubt. Nanoparticles are widely present in applied materials science and pharmaceutical technologies, and they often appear in fundamental research. However, even subtle changes in their polydispersity can significantly affect the efficiency, stability or safety of the relevant process or product. Consequently, gauging and controlling the particle-size distribution with respect to both the mean value and spread is of major importance for practical applications (Danaei *et al.*, 2018; Garcia *et al.*, 2009; Pastoriza-Gallego *et al.*, 2011; Schöpe *et al.*, 2007).

The use of small-angle scattering in the complex analysis of systems with nanoparticles provides insight into various physicochemical processes, since the method is able to give a precise structural description at the supramolecular level (Guinier & Fournet, 1955; Glatter & Kratky, 1982; Feigin & Svergun, 1987; Brumberger, 1995; Jeffries *et al.*, 2021). Performing experiments on the scattering of neutrons (small-angle neutron scattering, SANS) or X-rays (small-angle X-ray scattering, SAXS) together or separately allows one to analyze multi-phase systems in terms of scattering-length density, ρ. The growing popularity of *in situ* and *operando* structural analysis requires the development of methods for clear determination of the structure of materials at the mesoscale by scattering techniques (Peterson & Papadakis, 2015).

In the case of homogeneous monodisperse particles, through scattering-law analysis one can estimate the shape of



a nanoobject and its size, density and concentration. Polydispersity significantly smears the scattering curve, which is the dependence of the differential scattering cross section per unit sample volume, $I(q)$, on the modulus of the scattering vector, q (Guinier & Fournet, 1955).

Relatively high polydispersity causes the shape of the small-angle scattering curve of a system of particles to consist of a Gaussian-like Guinier regime at low q, a Porod power-law asymptote at high q and a narrow crossover region between them (Fig. 1). Beaucage proposed a unified exponential/power-law model to describe the scattering over the entire q range (Beaucage, 1995; Beaucage *et al.*, 2004):

$$I(q) = G\exp\left(-\frac{q^2R_{\rm g}^2}{3}\right) + Bq^{-4}\left[{\rm erf}\left(\frac{qR_{\rm g}}{\sqrt{6}}\right)\right]^{12}, \qquad (1)$$

where G is the forward scattering intensity, $R_{\rm g}$ is the gyration radius and B is the Porod prefactor. Numerous examples of applying this approach to various systems have been given in the literature (Beaucage *et al.*, 2004; Tomchuk *et al.*, 2014, 2021, 2022; Hammons *et al.*, 2018; Lázár *et al.*, 2020; Jeffries *et al.*, 2021; Zhang *et al.*, 2022). Although other form-factor models, such as Hammouda's (2010), are suitable for polydisperse systems, this study will concentrate on equation (1) due to its clear and informative representation. This model relates to the system being diluted enough to ignore the structure-factor effect, *i.e.* with negligible inter-particle correlations. Nevertheless, in some cases, such as aggregation, the structural level of particles can be described within the framework of the same formalism but using a multilevel unified scattering model (Beaucage, 1995).

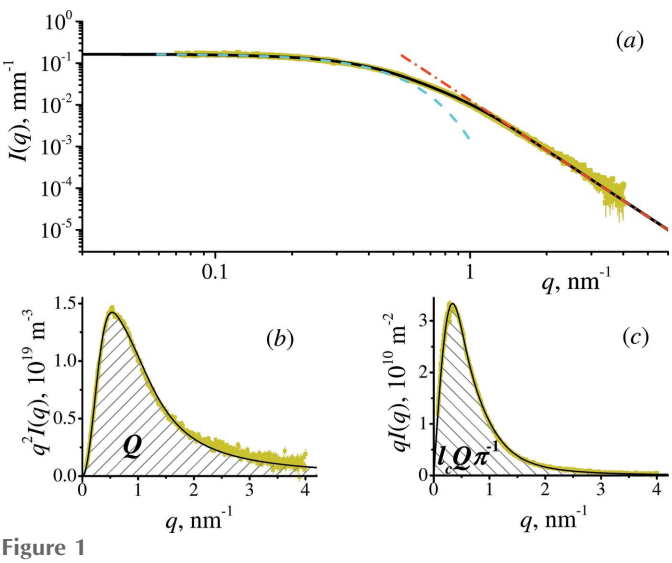


Figure 1
A typical SAXS curve from polydisperse gold nanoparticles dispersed in water (points with error bars) in (*a*) $I(q)$ versus q, (*b*) $q^2I(q)$ versus q and (*c*) $qI(q)$ versus q representation. Solid lines show the unified exponential/power-law approximation [equation (1)]. The dashed line shows the Guinier regime, $\sim\exp(-q^2R_{\rm g}^2/3)$, and the dash-dotted line corresponds to the Porod asymptote, $\sim q^{-4}$. Additionally, the definitions of the Porod integral, Q, and the correlation length, $l_{\rm c}$, according to equations (8*a*) and (8*b*), respectively, are visualized by hatching.

The particles can often be considered spherical. For example, for polyhedrons with gyration radius $R_{\rm g}$, the spherical approximation is valid for $qR_{\rm g} < 8$ (Feigin & Svergun, 1987), which is usually the case in an experiment. If this is the case, the scattering system is fully described by a distribution function of one variable, the radius R, as $D_{\rm N}(R)$.

To quantify polydispersity, the polydispersity index is used, and several definitions of the index exist. In dynamic light scattering (DLS), transmission electron microscopy (TEM) and small-angle scattering communities, the polydispersity index is usually defined by the ratio of the standard deviation, $\sigma = \sqrt{\langle R^2\rangle - \langle R\rangle^2}$, to the mean radius, $\langle R\rangle$:

$$p = \frac{\sigma}{\langle R\rangle}, \qquad (2)$$

where $\langle\ldots\rangle$ means averaging with the particle-size distribution function, $D_{\rm N}(R)$.

In polymer science, the polydispersity index is defined as mass-average molecular weight divided by number-average molecular weight:

$$\mathrm{PI} = \frac{M_{\rm w}}{M_{\rm n}} = \frac{\langle M^2\rangle}{\langle M\rangle^2}. \qquad (3)$$

A shape-independent definition of the polydispersity index based on experimental features of small-angle scattering patterns was given by Beaucage *et al.* (2004):

$$\mathrm{PDI} = \frac{50}{81}\frac{BR_{\rm g}^4}{G}. \qquad (4)$$

In order to avoid confusion as to what polydispersity index means in a particular context, we restrict the use of the PDI abbreviation to the latter sense only, *i.e.* as defined by Beaucage *et al.* (2004). The $\sigma/\langle R\rangle$ quantity will be denoted by p and the polymer-science mass-based definition by PI. The values of such defined polydispersity indices are compared in Fig. 2 for several lognormal distributions of different widths.

The p parameter seems to be the most interdisciplinary, and therefore the most frequently referred to. Small-angle scat-

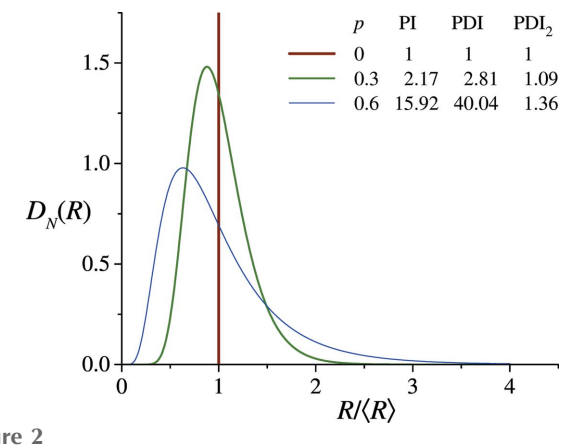


Figure 2
Lognormal size distributions of different widths. The inset shows the corresponding polydispersity indices discussed: p [equation (2)], PI [equation (3)], PDI [equation (4)] and PDI_2 [equation (21)].

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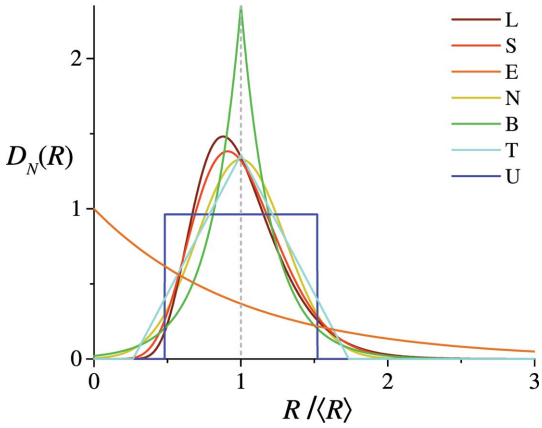

Figure 3
Particle-size distributions including lognormal (L), Schulz (S), exponential (E), normal (N), Boltzmann (B), triangular (T) and uniform (U) functions. For all of them, the polydispersity was $p = 0.3$, except for the exponential one, where, by definition, $p = 1$.

tering is the most versatile method to characterize polydispersity since it does not have the limitations of TEM (small statistics) and DLS (solution transparency, low concentration). In addition, only small-angle scattering can work with porous systems in a non-destructive way (Brumberger, 1995). Consequently, the study of the relationship between small-angle scattering based PDI and p is especially useful.

Each of the parameters given in equation (1) is proportional to some non-central moment of the size distribution function, $\langle R^k\rangle$ (see details below in Section 2.1). This formed the basis of an approach that makes it possible to analytically calculate the parameters of the lognormal size distribution according to a dimensionless combination [equation (4)] (Beaucage *et al.*, 2004). Later, this approach was developed for the case of inhomogeneous particles characterized by a certain type of diffusive surface (Tomchuk *et al.*, 2014). This is in contrast to another approach where $I(q)$ is considered as the sum of monodisperse scattering functions weighted according to $D_{\rm N}(R)$ fitted against the scattering data by the maximum entropy method (Potton *et al.*, 1988; Ilavsky & Jemian, 2009) or conventional regularization provided by the known function shape (Glatter, 1980; Glatter & Kratky, 1982; Pedersen, 1997). A significant advantage of the PDI method is the absence of the integral equations in the approach, which could otherwise be a potential source of additional mathematical artefacts in the resulting size distribution function and in the subsequent determination of p.

Nevertheless, the lognormal function is far from the only one encountered in practice when working with advanced nanosystems. This article discusses the possibilities of obtaining polydispersity parameters, $\langle R\rangle$ and p, for some other common $D_{\rm N}(R)$ types (Fig. 3). The PDI-based approach seems to be useful for fast and unambiguous estimation of polydispersity parameters in small-angle scattering experiments. In addition, the question of whether it is possible to suggest the type of $D_{\rm N}(R)$ functional dependence solely from small-angle scattering data is considered in detail. Furthermore, experimental SAXS data obtained for hydrosols of gold, silver and copper nanoparticles are used to examine the possibilities of the proposed approach.

2. Theoretical development of the PDI formalism

2.1. Relation between scattering parameters and moments of the size distribution function of spherical particles

For a two-phase particle–solvent system, the scattering is proportional to the product of the volume fractions of the two phases, $\varphi(1-\varphi)$. The consideration of sufficiently dilute systems of nanoparticles, in addition to excluding the effect of the structure factor indicated above, has another important consequence. The volume fraction of one of the phases tends towards 1. In such an approximation, the intensity of small-angle scattering is proportional to the volume fraction of particles, φ, and, consequently, the number density, n (Feigin & Svergun, 1987). Thus, some of the parameters of the scattering curve considered below are also proportional to the number density. This means that it can be canceled in ratios, which was used in the study of Beaucage *et al.* (2004).

For a system of homogeneous particles placed in a solvent, the scattering parameters G, $R_{\rm g}$ and B are related to non-central moments of the polydispersity function as (Brumberger, 1995; Feigin & Svergun, 1987)

$$G = n(\Delta\rho)^2\left(\frac{4\pi}{3}\right)^2\langle R^6\rangle, \qquad (5a)$$

$$R_{\rm g} = \sqrt{\frac{3\langle R^8\rangle}{5\langle R^6\rangle}} \qquad (5b)$$

and

$$B = n(\Delta\rho)^2 8\pi^2\langle R^2\rangle. \qquad (5c)$$

The contrast $\Delta\rho$ is defined as the difference between the scattering-length densities of the particles, $\rho_{\rm part}$, and of the solvent, $\rho_{\rm solv}$, $\Delta\rho = \rho_{\rm part} - \rho_{\rm solv}$, which for X-rays is proportional to the difference in electron density (away from absorption edges).

In the general structural analysis, the lower-order moments of the size distribution are represented by scattering invariants. Thus, the Porod invariant is defined as the integral of a Kratky-type plot (Glatter & Kratky, 1982; Brumberger, 1995),

$$Q = \int_0^\infty q^2 I(q)\,{\rm d}q = n(\Delta\rho)^2\frac{8\pi^3}{3}\langle R^3\rangle, \qquad (6)$$

and is proportional to $\langle R^3\rangle$. The correlation length, *i.e.* weighted average chord length (Glatter & Kratky, 1982), $l_{\rm c}$, contains information about $\langle R^4\rangle$ (Möller *et al.*, 1995):

$$l_{\rm c} = \frac{\pi}{Q}\int_0^\infty qI(q)\,{\rm d}q = \frac{3}{2}\frac{\langle R^4\rangle}{\langle R^3\rangle}. \qquad (7)$$

Often, due to the integration of experimental data in equations (6) and (7), the errors in determining Q and $l_{\rm c}$ are quite large. In addition, the experimental q range of existing

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instruments is typically limited to two orders of magnitude, which can also have an effect. Alternatively, the functional form of the scattering law [equation (1)] makes it possible to obtain expressions for these scattering invariants analytically, as functions of G, R_g and B only, with no need to integrate the experimental data:

$$Q = \frac{3\sqrt{3\pi}\,G}{4R_g^3} + \frac{BR_gC_1}{\sqrt{6}} \tag{8a}$$

and

$$l_c = \frac{3\pi G}{2R_g^2 Q} + \frac{\pi B R_g^2 C_2}{6Q}, \tag{8b}$$

where

$$C_1 = \int_0^\infty x^{-2}[\mathrm{erf}(x)]^{12}\,\mathrm{d}x \simeq 0.771965$$

and

$$C_2 = \int_0^\infty x^{-3}[\mathrm{erf}(x)]^{12}\,\mathrm{d}x \simeq 0.319739.$$

The quantities Q and $l_cQ\pi^{-1}$ were calculated from the $I(q)$ pattern shown in Fig. 1(a), and are presented in Figs. 1(b) and 1(c) in the form of hatched areas underneath the $q^2I(q)$ and $qI(q)$ curves, respectively. Both Q and $l_cQ\pi^{-1}$ appear as good estimates of the integrals present in equations (6) and (7), as expected.

2.2. PDI method with a known distribution type

As stated above, Beaucage *et al.* (2004) proposed a dimensionless parameter, PDI, to compare the polydispersity of different systems [equation (4)]. Taking into account equations (5a), (5b) and (5c), it is actually the combination of moments of the size distribution function (Beaucage *et al.*, 2004):

$$\mathrm{PDI} = \frac{\langle R^2\rangle\langle R^8\rangle^2}{\langle R^6\rangle^3}. \tag{9}$$

Obviously, $\lim_{p\to 0}\mathrm{PDI} = 1$. PDI is directly related to p and increases rapidly with increasing polydispersity, since it contains distribution moments up to the eighth order.

A partial solution for the lognormal distribution type is obtained when the size distribution parameters are analytical functions of G, R_g and B (Beaucage *et al.*, 2004). The lognormal function belongs to the family of continuous two-parameter distributions:

$$D_N(R) = \frac{1}{Rs\sqrt{2\pi}}\exp\left\{-\frac{[\ln(R/R_0)]^2}{2s^2}\right\}. \tag{10}$$

Its parameters can be expressed through the mean radius and polydispersity:

$$s = \sqrt{\ln(1+p^2)} \tag{11a}$$

and

$$R_0 = \frac{\langle R\rangle}{\sqrt{1+p^2}}. \tag{11b}$$

All moments of the lognormal size distribution can be calculated (Table 1), which makes it possible to obtain not only the direct analytical dependence,

$$\mathrm{PDI} = \left(1+p^2\right)^{12}, \tag{12}$$

but also the inverse one,

$$p = \sqrt{\mathrm{PDI}^{1/12} - 1}. \tag{13}$$

Beaucage *et al.* (2004) obtained the equations for both parameters of the lognormal distribution:

$$s = \sqrt{\frac{1}{12}\ln(\mathrm{PDI})} \tag{14a}$$

and

$$R_0 = R_g\sqrt{\frac{5}{3\exp(14s^2)}}. \tag{14b}$$

Equations (14a) and (14b) are known as the PDI–R_g method, which has become popular in recent times (Tomchuk *et al.*, 2021, 2022; Michelsen *et al.*, 2022; Ilavsky & Jemian, 2009). As far as the analytical determination of polydispersity parameters from scattering data is concerned, the lognormal distribution is rather an exception to the rule. Nevertheless, it is always possible to obtain direct analytical dependences PDI = PDI(p) by equation (9) for further calculation of the inverse functions $p = p$(PDI) with arbitrarily high precision (Fig. 4). The latter opens the way to determining the p value from the fitting parameters of the unified exponential/power-law model [equation (1)]. Therefore, we considered several size distribution functions that are most commonly used in practice (Fig. 3).

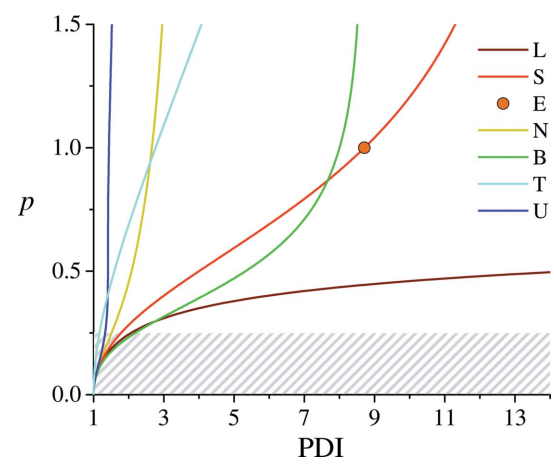

Figure 4
The dependences of the p parameter on the PDI index for various size distributions according to Table 1 are presented. The gray hatched area indicates the limitations of polydispersity analysis by PDI at the interval $p < 0.25$. The designations of the dependences are the same as in Fig. 3.

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Table 1
Equations for the determination of polydispersity parameters, p and $\langle R\rangle$, based on the unified exponential/power-law approach via PDI and $R_{\rm g}$, using equations (4), (5*a*), (5*b*), (5*c*) and (9).

Distribution	kth moment, $\langle R^k\rangle$	PDI	Mean radius, $\langle R\rangle$
Lognormal	$\langle R\rangle^k\left(1+p^2\right)^{k(k-1)/2}$	$\left(1+p^2\right)^{12}$	$\sqrt{\frac{5}{3}}R_{\rm g}\left(1+p^2\right)^{-13/2}$
Exponential	$\frac{k!}{\lambda^k}$	8.7111	$\sqrt{\frac{5}{3}}R_{\rm g}\frac{1}{2\sqrt{14}}$
Schulz	$\langle R\rangle^k\frac{(z+k)!}{(z+1)^k z!}$	$\frac{\langle R^2\rangle\langle R^8\rangle^2}{\langle R^6\rangle^3}$	$\sqrt{\frac{5}{3}}R_{\rm g}\frac{z+1}{\sqrt{(z+8)(z+7)}}$
Normal	$\langle R\rangle^k\left(1+A_2p^2+3A_4p^4+15A_6p^6+\cdots\right)$ $A_m=\frac{k!}{(k-m)!m!}$	$\frac{\langle R^2\rangle\langle R^8\rangle^2}{\langle R^6\rangle^3}$	$\sqrt{\frac{5}{3}}R_{\rm g}\sqrt{\frac{1+15p^2+45p^4+15p^6}{1+28p^2+210p^4+420p^6+105p^8}}$
Boltzmann	$\sum_{i=0}^{\lfloor k/2\rfloor}\frac{\langle R\rangle^{k-2i}\sigma^{2i}}{2^i}\frac{k!}{k-2i}$	$\frac{\langle R^2\rangle\langle R^8\rangle^2}{\langle R^6\rangle^3}$	$\sqrt{\frac{5}{3}}R_{\rm g}\sqrt{\frac{1+15p^2+90p^4+90p^6}{1+28p^2+420p^4+2520p^6+2520p^8}}$
Triangular	$\frac{\left(\langle R\rangle-\sigma\sqrt{6}\right)^{k+2}+\left(\langle R\rangle+\sigma\sqrt{6}\right)^{k+2}-2\langle R\rangle^{k+2}}{6\sigma^2(k+1)(k+2)}$	$\frac{\langle R^2\rangle\langle R^8\rangle^2}{\langle R^6\rangle^3}$	$\sqrt{\frac{5}{3}}R_{\rm g}\sqrt{\frac{1+\frac{5}{2}p^2+p^4+\frac{1}{28}p^6}{1+\frac{14}{3}p^2+\frac{14}{3}p^4+p^6+\frac{1}{45}p^8}}$
Uniform	$\frac{\left(\langle R\rangle+\sigma\sqrt{3}\right)^k-\left(\langle R\rangle-\sigma\sqrt{3}\right)^k}{\sigma 2\sqrt{3}(k+1)}$	$\frac{\langle R^2\rangle\langle R^8\rangle^2}{\langle R^6\rangle^3}$	$\sqrt{\frac{5}{3}}R_{\rm g}\sqrt{\frac{1+15p^2+27p^4+\frac{27}{7}p^6}{1+28p^2+126p^4+108p^6+9p^8}}$

The next example of an asymmetric two-parameter distribution is the Schulz distribution,

$$D_{\rm N}(R)=(z+1)^{z+1}\left(\frac{R}{\langle R\rangle}\right)^z\frac{\exp[-(z+1)R/\langle R\rangle]}{\langle R\rangle\Gamma(z+1)},\qquad(15)$$

where $z=1\big/p^2-1$ and $\Gamma(x)$ is the Euler gamma function. The functional dependence is close to the lognormal law, but with somewhat weighty tails (Kotlarchyk *et al.*, 1988). It is widely used for various systems, from micro-emulsions to magnetic nanoparticles (Nagornyi *et al.*, 2020).

For completeness, we also consider the exponential distribution, widely used in polymer science,

$$D_{\rm N}(R)=\lambda\exp(-\lambda R).\qquad(16)$$

It somewhat drops out of our research because it has one parameter, λ. Consequently, by excluding one distribution parameter from the PDI definition, we get a single value of PDI = 8.7111, rather than a function, which is close to the previously reported 8.70 for the most probable distribution (Beaucage *et al.*, 2004). Mathematically, the exponential distribution is just a subcase under p = 1 of the Schulz distribution [equation (15)] described above.

Returning to two-parameter continuous distributions, we next consider symmetric polydispersity functions. The normal (Gaussian) size distribution is the most widespread and is included in many data-processing packages since its parameters are explicitly related to the mean radius, $\langle R\rangle$, and standard deviation, σ:

$$D_{\rm N}(R)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{(R-\langle R\rangle)^2}{2\sigma^2}\right].\qquad(17)$$

The second-order exponential gives fast decay at the tails. A first-order exponential function is represented here by the Boltzmann (Laplace) distribution,

$$D_{\rm N}(R)=\frac{1}{\sigma\sqrt{2}}\exp\left(-\frac{\sqrt{2}}{\sigma}|R-\langle R\rangle|\right),\qquad(18)$$

which has a sharper peak in the vicinity of $\langle R\rangle$ but slower decay at $R\to 0$ and $R\to\infty$.

In addition, somewhat idealized cases of symmetric triangular (Simpson),

$$D_{\rm N}(R)=\begin{cases}\frac{1}{\sigma\sqrt{6}}-\frac{1}{6\sigma^2}|\langle R\rangle-R|, & \\ \qquad\left(\langle R\rangle-\sigma\sqrt{6}\right)<R<\left(\langle R\rangle+\sigma\sqrt{6}\right), & \\ 0, \quad R<\left(\langle R\rangle-\sigma\sqrt{6}\right),\ R>\left(\langle R\rangle+\sigma\sqrt{6}\right), & \end{cases}\qquad(19)$$

and uniform (rectangular),

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$$D_{\rm N}(R) = \begin{cases} \dfrac{1}{\sigma 2\sqrt{3}}, & \left(\langle R\rangle - \sigma\sqrt{3}\right) < R < \left(\langle R\rangle + \sigma\sqrt{3}\right), \\ 0, & R < \left(\langle R\rangle - \sigma\sqrt{3}\right),\ R > \left(\langle R\rangle + \sigma\sqrt{3}\right), \end{cases} \tag{20}$$

distributions were also analyzed.

For each of the presented distributions (Fig. 3), the p versus PDI dependence has a special form (Fig. 4). The calculations were made according to the formulae collected in Table 1. *A priori* information about the distribution type, obtained using other methods or derived from the characteristics of the synthesis process, is important in determining the polydispersity in terms of p. Otherwise the results can differ significantly for different $D_{\rm N}(R)$ functions.

Consequently, it is possible to derive the mean radius of nanoparticles, $\langle R\rangle$, with the help of the equations given in Table 1, using the p parameter, as well as the gyration radius, $R_{\rm g}$.

By analogy to the previously developed method for the lognormal distribution, it becomes possible to obtain the polydispersity parameters of a system of particles from small-angle scattering data for various size distribution types.

Fig. 4 indicates a certain lower limit for the parameter p, since, at such a low polydispersity, the small-angle scattering curve will not be satisfactorily described by means of Beaucage's unified model. Form-factor oscillations will also appear due to the high symmetry of sphere-like particles. Both of these aspects lead to the conclusion that the approach presented in this article should be exclusively applied to highly polydispersed systems. The conventional limit should be set at the level of p = 0.25. Nonetheless, it may differ slightly for different size distribution functions.

2.3. Analysis of polydispersity in the case of an unknown distribution type

Occasionally, preliminary information about the sample structure is poor concerning polydispersity. Therefore, it is valuable to have methods for determining the type of size distribution function based solely on the experimental scattering data. For this purpose, it is proposed to use an approach that is a continuation of the ideas outlined in the previous section. Since the PDI contains moments of the size distribution of high orders (up to eighth), then let us, by analogy to equations (4) and (9), introduce a dimensionless combination of scattering parameters determined by lower orders, from the second to the fourth,

$$\mathrm{PDI}_2 = \frac{2\pi}{9}\frac{Bl_{\rm c}}{Q} = \frac{\langle R^2\rangle\langle R^4\rangle}{\langle R^3\rangle^2}. \tag{21}$$

Again, it is normalized to unity, $\lim_{p\to 0}\mathrm{PDI}_2 = 1$. Incidentally, similarly to PDI, PDI_2 can be used for experimental data presented in arbitrary units, *i.e.* normalization to the sample thickness is optional. Here, Q and $l_{\rm c}$ can be obtained analytically using equations (8*a*) and (8*b*), respectively, to decrease the PDI_2 errors. An example of using PDI_2 to characterize the polydispersity of spherical nanoparticles is shown in Fig. 2, along with other polydispersity indices.

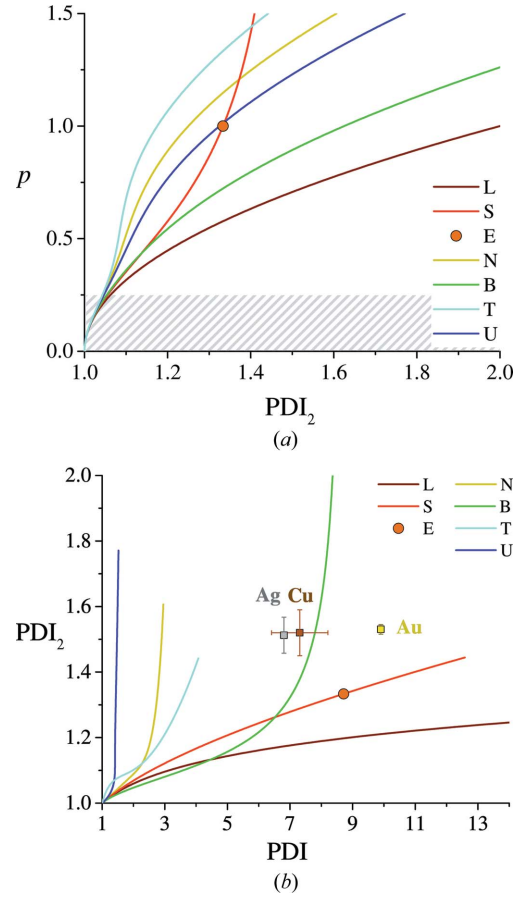

Figure 5
(*a*) p versus PDI_2 dependences for various size distributions (labeled as in Fig. 3). Again, the gray hatched area indicates the minimal sensitivity limit by p. (*b*) A proposed PDI_2 versus PDI diagram for the analysis of the particles' polydispersity function as p-parameterized plots. Curves and a full circle demonstrate theoretical predictions, and points with error bars show experimental results for the three hydrosols under study.

Fig. 5(*a*) indicates that PDI_2 changes less drastically with increasing polydispersity (p growth). The sensitivity limit p = 0.25 can be suggested in this case as well. However, the nature of these changes differs from those for the classical PDI (Fig. 4). This gives grounds for the analysis of the scattering data based on both dimensionless parameters simultaneously [Fig. 5(*b*)]. The values of both quantities can indicate the proximity to certain distributions in the PDI versus PDI_2 coordinates. The physical meaning is that both indices correspond to the same value of the p parameter. This method allows the type of size distribution function to be determined. The mean size and polydispersity, namely $\langle R\rangle$ and p, can then be calculated using the formalism described in the previous section.

3. Experiment: metallic nanoparticle hydrosols

To verify the developed approach, experimental SAXS data from aqueous sols of polydisperse metal particles were examined.

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TEM images of dried samples of the studied silver, gold and copper hydrosols are presented in Fig. 6, together with calculated size distributions. Silver nanoparticles form an agglomeration-like pattern in contrast to gold and copper nanoparticles. The aggregation of nanoparticles could result from the drying stage in the preparation procedure (Shrestha *et al.*, 2020). However, the polyvinylpyrrolidone coating prevents particle contact and allows fairly accurate sizing.

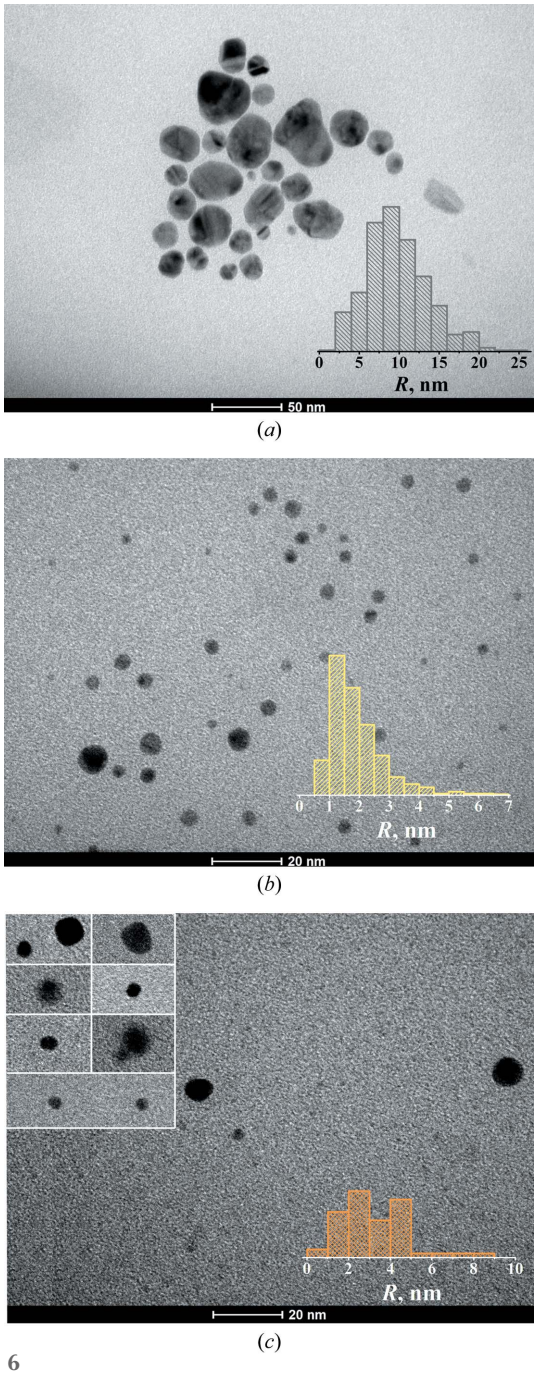

Figure 6
Bright-field TEM images of (*a*) silver, (*b*) gold and (*c*) copper nanoparticles on an amorphous carbon support. Particle-size distribution histograms are shown in the insets. The insets in the upper left corner of (*c*) show copper nanoparticles distributed out of the main image, shown here with the same scalebar, as there were not enough particles observed in one area of the image.

Table 2
Structural parameters of the studied metallic nanoparticles according to small-angle scattering data within the framework of the proposed PDI/PDI_2 formalism.

Nanoparticle type	Ag	Au	Cu
G (mm^{-1})	11.80 (4)	0.1647 (5)	0.00346 (4)
R_g (nm)	15.12 (4)	3.78 (1)	1.64 (6)
B (mm^{-1} nm^{-4})	0.00249 (2)	0.01298 (4)	0.00563 (6)
Q (mm^{-1} nm^{-3})	0.0197 (1)	0.02248 (6)	0.0047 (1)
l_c (nm)	17.16 (6)	3.80 (3)	1.82 (7)
PDI	6.8 (1)	9.9 (1)	7.3 (9)
PDI_2	1.51 (5)	1.53 (1)	1.52 (7)
$D_N(R)$	Boltzmann	Schulz	Boltzmann
p	0.68 (2)	1.04 (2)	0.77 (4)
$\langle R \rangle$ (nm)	5.45 (5)	0.61 (3)	0.52 (7)

In the solution, nanoparticles are considered to be non-interacting because of their low concentration. They are also considered to be homogeneous from the SAXS viewpoint as a result of the much higher contrast of metal against water in comparison with that for polyvinylpyrrolidone stabilizer.

Experimental background-corrected SAXS curves together with best fits of the model of the unified exponential/power-law approach [equation (1)] are given in Fig. 7. The fitting parameters, as well as the results of the polydispersity analysis within the proposed approach, are given in Table 2. First, the scattering integrals Q and l_c were determined. Then, the PDI and PDI_2 parameters were calculated, which were further compared with the theory in Fig. 5(*b*). As one can see, the experimental points are far from the line corresponding to the lognormal distribution. This means that it would be incorrect to use the PDI analysis method in the lognormal approximation. The more appropriate description developed in the previous sections should be used instead. Of course, we do not have a polydispersity function that is exactly described by one of the considered distributions. Still, one can judge the degree of closeness of the experiment to one or another $D_N(R)$ type. Thus, we concluded that the Boltzmann distribution was the

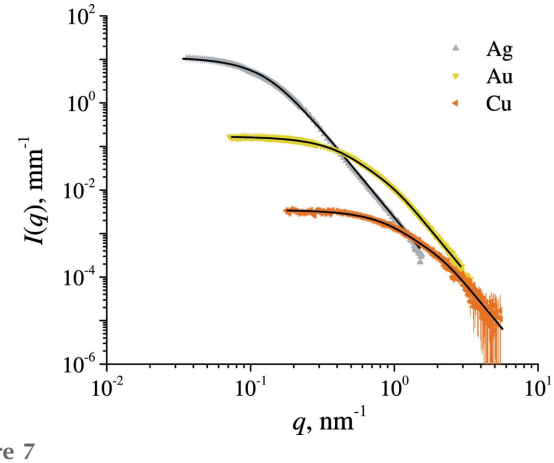

Figure 7
Experimental SAXS curves (points) for three investigated aqueous sols of nanoparticles (indicated in the legend). Model solid lines correspond to the unified exponential/power-law approximation [equation (1)]. Corresponding results of fitting data and calculating polydispersity parameters are presented in Table 2.

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most appropriate for silver and copper particles, while the case of gold particles tends to the Schulz distribution. After determining the type of polydispersity function, the average size and polydispersity parameter p were calculated according to the equations in Table 1.

The results show extremely wide size distributions (Table 2). Thus, we can conclude that there is a significant fraction of nanometre and subnanometre particles, which is, in principle, characteristic of particles synthesized according to chemical bottom-up schemes (de Oliveira *et al.*, 2020; Betke & Kickelbick, 2014). Relatively low mean radius values according to SAXS data indicate this feature. Somewhat larger sizes according to TEM data may indicate that the observed dimensions were at the lower limit of sensitivity of the electron microscope used. The statistical significance here is in favor of SAXS since the information is obtained by scattering on billions of particles in a hydrosol sample volume of the order of $\sim 0.5 \times 0.5 \times 1.5$ mm compared with several hundred particles (or even several tens for the copper case) studied from TEM micrographs. However, the overall trend in the size ratio of the three types of particles was the same for both techniques used. Namely, comparable sizes of copper and gold particles versus moderately larger silver particles were observed.

Therefore, the proposed approach can be successfully applied to the analysis of the polydispersity of a broad class of highly polydisperse nanosystems including porous systems.

4. Conclusions

Particle-size polydispersity is a common case in practical applications. A generalized approach to the analysis of particle polydispersity, given the small-angle scattering data, was proposed. The presented theory is a natural development of ideas presented in the literature for almost two decades (Beaucage *et al.*, 2004). In this article, we considered several two-parameter size distributions whose parameters can be uniquely obtained from the parameters of the unified exponential/power-law approximation to small-angle neutron or X-ray scattering.

Moreover, by analogy with the PDI, we introduced another dimensionless parameter composed of lower-order moments of the size distribution, PDI_2. Simultaneous consideration of both quantities in certain cases makes it possible draw conclusions about the type of size distribution exclusively from small-angle scattering data. The developed approach was successfully demonstrated by the structural analysis of aqueous sols of silver, gold and copper nanoparticles. The proposed methodology for analyzing scattering data can be represented as the following sequential steps:

Step (1). Fitting the data with equation (1) to obtain R_g, B and G.

Step (2). Determination of PDI from equation (4).

Step (3). Calculation of the invariants Q and l_c using equations (6) and (7), respectively.

Step (4). Determination of PDI_2 from equation (21).

Step (5). Using the PDI–PDI_2 diagram [Fig. 5(*b*)] to determine the type of two-parameter distribution.

Step (6). Determination of the polydispersity parameter p according to the PDI–p diagram (Fig. 4 or Table 1).

Step (7). Calculation of the average size, $\langle R \rangle$, according to Table 1.

In summary, on the basis of only three fitting parameters of the unified function [equation (3)], we can draw conclusions about the type of size distribution function. The presented approach to small-angle scattering data analysis gives fast and unambiguous estimates of the mean size and distribution width for the most common polydispersity functions. This kind of analysis of SANS and SAXS data has excellent prospects for automation, and is devoid of a series of shortcomings of $D_N(R)$ determination techniques based on minimizing χ^2 or maximizing entropy relative to experimental data.

5. Materials and methods

In this study, silver, gold and copper quasi-spherical nanoparticles stabilized in an aqueous medium with cation-type polymer polyvinylpyrrolidone were purchased from M9, Tolyatti. According to DLS data obtained with a Photocor Compact-Z analyzer in the intensity mode (Photocor Instruments, Tallinn), their average hydrodynamic radius was (18.2 ± 2.0) nm (Ag), (10.4 ± 6.7) nm (Au) and (6.8 ± 1.2) nm (Cu), respectively. Before measurements, the solution was passed through a 0.45 µm filter. For each of the three studied hydrosols, the concentration was 0.02 wt%.

A Tecnai G2 20 TWIN (FEI, Hillsboro) transmission electron microscope with a LaB_6 cathode operated at an accelerating voltage of 200 kV was used. A dozen bright-field TEM images from each specimen were acquired to improve statistics. The particle sizes were determined manually using the *ImageJ* software (Schneider *et al.*, 2012). Two axes were used in each case (the axis of maximum size and perpendicular to it) to eliminate the effect of anisotropy present for some fraction of particles of each type. The average particle radii were 9.7 nm (Ag), 1.9 nm (Au) and 3.2 nm (Cu), calculated with 494, 1108 and 56 particles for silver, gold and copper, respectively.

The SAXS experiments were carried out on a Xeuss 3.0 instrument (Xenocs, Grenoble) with a Cu GeniX^{3D} source (working X-ray wavelength of 0.154 nm). The high-resolution configuration was operated. A silver behenate standard for distance calibration and a glassy carbon standard for absolute intensity calibration were used. The studied hydrosols were placed into 1.5 mm borosilicon capillaries (Hilgenberg GmbH). Data treatment was carried out using the *SasView 5.0.5* software (https://www.sasview.org/) with the built-in scattering form factors.

Acknowledgements

The authors appreciate discussions with Leonid Bulavin (Taras Shevchenko National University of Kyiv, Ukraine) and

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